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WORK IS A REAL KILLER

How the economy influences traffic fatalities

Hypothesis

When unemployment rises, fewer people drive to work, leading to fewer traffic fatalities overall

Variable	Description
totfatrte	total traffic fatality rate (per 100,000 population)
unem	state unemployment rate (%)
zerotol	zero tolerance underage DUI law indicator
minage	minimum legal drinking age indicator
bac08	BAC 0.08 law indicator
bac10	BAC 0.10 law indicator
wkndfatrte	weekend traffic fatality rate (per 100,000 population)

*All models have fixed effects for state and robust standard errors for heteroskedasticity

Weekend Fatalities: Model 2

$$wkndfatrte_i = \beta_1 + \beta_2 unem_i + \beta_3 zerotol_i + \beta_4 minage_i + \beta_5 bac08_i + \beta_6 bac10_i + \varepsilon_i$$

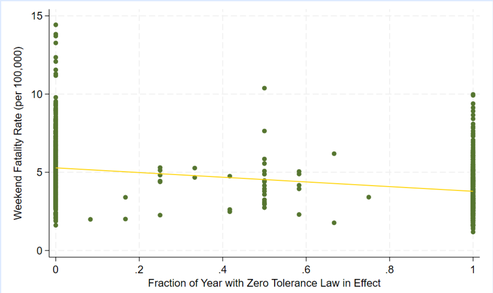
Variable	Coefficient	Robust Standard Errors	T-value
unem	-0.8322	0.0302	-2.75
zerotol	-0.9365	0.0928	-10.09
minage	-0.3529	0.0774	-4.56
bac08	-1.7642	0.3788	-4.66
bac10	-1.2392	0.2727	-4.54
constant	13.9419	1.5387	9.06

- Alcohol-related policies significantly reduce weekend traffic fatalities
- The relationship between unemployment and weekend fatalities is weaker and less precisely estimated.

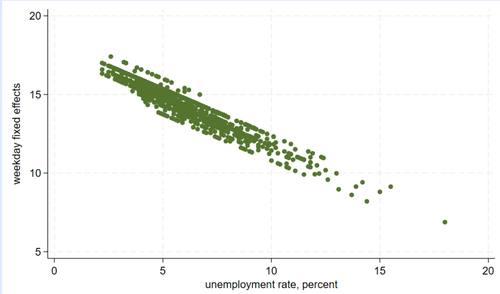
Model 1

$$totfatrte_i = \beta_1 + \beta_2 unem_i + \beta_3 zerotol_i + \beta_4 minage_i + \beta_5 bac08_i + \beta_6 bac10_i + \varepsilon_i$$

Variable	Coefficient	Robust Standard Error	T-value
unem	-0.4522	0.0923	-4.9
zerotol	-2.1616	0.3914	-5.52
minage	-0.7132	0.2376	-3
bac08	-4.7147	0.9979	-4.72
bac10	-3.1615	0.7415	-4.26
constant	40.253	4.9972	8.06



Model 2



Model 3

Conclusion

- Weekday fatalities show a strong and statistically significant relationship with economic conditions.
- Alcohol-related policies show a strong and statistically significant relationship with weekend fatalities.
- Economic conditions cannot explain all fatalities

So what does this mean?

- Higher unemployment is linked to fewer total traffic fatalities
- Alcohol-related policy variables are consistently negative and statistically significant, even with fixed effects and robust standard errors.
- Weekday and weekend driving patterns differ, what happens when we separate the two?

Weekday Fatalities: Model 3

wkdayfatrte=totfatrte-wkndfatrte

$$wkdayfatrte_i = \beta_1 + \beta_2 unem_i + \beta_3 zerotol_i + \beta_4 minage_i + \beta_5 bac08_i + \beta_6 bac10_i + \varepsilon_i$$

Variable	Coefficient	Robust Standard Errors	T-value
unem	-0.3690	0.0676	-5.46
zerotol	-1.2250	0.3180	-3.85
minage	-0.3603	0.1807	-1.99
bac08	-2.9506	0.6573	-4.49
bac10	-1.9222	0.5141	-3.74
constant	26.3110	3.8552	6.82

- Higher unemployment is associated with fewer weekday traffic fatalities
- Both unemployment and alcohol-related policies significantly affect weekday fatalities, with unemployment showing strong statistical significance.